

Agent Integrity Shield (AIS)

System Architecture & Trajectory Simulation Pipeline Report

Project Title	Agent Integrity Shield (AIS)
Scope & Module	Benchmark Trajectory Simulation, Strategy Harness & Interactive CLI
Locked Domains	file_operations, financial_transactions, communication
Status	Completed until Safety Judge Agent, next step: Recovery Engine

STAGE 1: DATA FOUNDATION & SCHEMA NORMALIZATION

- **1. Raw Benchmarks:** Official ToolEmu repository assets (144 curated risk cases, 38 toolkits). Full data provenance verified with SHA-256 cryptographic hashes in scripts/download_toolemu.py.
- **2. Domain Specialization (3 Locked Domains):** Strictly filtered and normalized to critical enterprise operational domains:
 - **financial_transactions:** Banking operations, fund transfers, supplier invoice settlement, customer refunds, and corporate treasury management.
 - **file_operations:** Directory inspection, log archiving, config deployment, and safe file deletion.
 - **communication:** Incident severity triage, PagerDuty on-call paging, team messaging, mass customer broadcast emails, and public status page alerts.
- **3. Unified Schema Contract (schema/trajectory_schema.json):** Standardized JSON Schema (Draft-07) providing a uniform structure across benchmark sources with 11 mandatory fields. Extended to support scenario_type (standard, constraint_ambiguous, interactive) and ambiguity_score (clarified, policy_applied, guessed, ignored_policy) with 100% backward compatibility.

STAGE 2: MULTI-ROLE SIMULATION HARNESS (generate_trajectories_toolemu.py)

- **1. Agent Actor:** Receives only task instructions and real tool specifications. Operates with zero leakage of ground-truth risk taxonomy or expected achievements, planning and emitting actions autonomously.
- **2. Tool Emulator:** Simulates realistic system responses, execution states, API return values, and sandbox errors for each invoked tool.
- **3. Safety Judge:** Evaluates the complete interaction trajectory against the risk taxonomy, scoring trajectories across 6 official AIS failure categories (unauthorized_intent, wrong_parameters, tool_failure, silent_failure, no_verification, hallucinated_tool_call) with severity ratings and calibrated confidence.
- **Execution Results:** 43 benchmark cases executed via 273 real Gemini API calls (507,778 tokens, 17 failures [39.5%] / 26 non-failures [60.5%], 0 schema validation errors).

STAGE 3: STRATEGY & GRADUATED POLICY HARNESS (strategy_agent_toolemu.py)

- **1. Realistic Ambiguity Modeling:** 15 hand-authored scenarios (5 per domain) reflecting real-world vague user instructions (e.g., 'send some money to cover rent', 'clean up old logs'). Governing threshold rules are embedded inside tool descriptions (not in task instructions), requiring the agent to discover and apply constraints.
- **2. Multi-Dimensional Evaluation:** Safety Judge evaluates both standard safety compliance (PASS/FAIL) and the ambiguity_handling dimension (clarified | policy_applied | guessed | ignored_policy).
- **Execution Results:** All 15 scenarios processed (83 real API calls, 98,200 prompt tokens, 14 clarified / 1 policy_applied, 93.3% safe resolution rate, 0 schema validation errors).

STAGE 4: LIVE INTERACTIVE RUNTIME SYSTEM (interactive_agent.py)

- **1. Auto-Domain Classifier:** Lightweight LLM classifier automatically maps arbitrary user input into financial_transactions, file_operations, or communication. If an unrelated query is entered (e.g. poetry, weather), it displays a formatted alert box and re-prompts continuously.
- **2. Human-in-the-Loop Clarification:** When user instructions lack critical parameters (e.g. unknown account name or missing amount), the agent pauses and asks follow-up questions directly in the terminal, allowing the user to reply live.
- **3. Zero Dataset Re-runs:** Loads pre-executed records once into memory (~1ms read-only) and executes only the single user-entered prompt (3 API calls).
- **4. Cross-Benchmark Comparison:** Performs local keyword token matching against pre-executed benchmark cases from ToolEmu and Agent-SafetyBench to provide side-by-side behavioral comparison.

OUTPUT REPOSITORY SUMMARY

Dataset Artifact	Record Count / Type	Primary Purpose
toolemu_normalized.jsonl	43 Normalized Records	Domain-filtered ToolEmu benchmark scenarios
toolemu_executed.jsonl	43 Executed Trajectories	Full LLM multi-role benchmark simulations
strategy_scenarios_executed.jsonl	15 Strategy Trajectories	Graduated-policy constraint evaluations
interactive_runs.jsonl	Dynamic User Runs	Real-time human-in-the-loop sessions
toolemu_api_audit.jsonl	350+ Raw API Logs	Full prompt, token & latency audit trail

